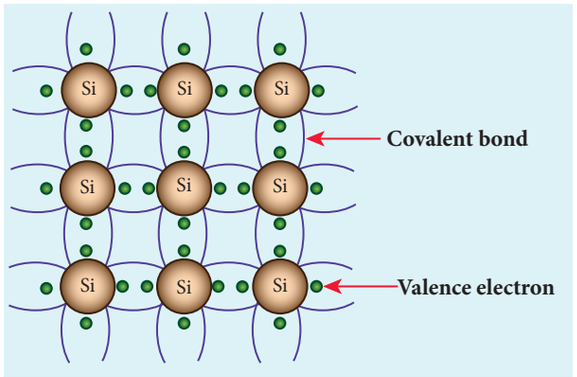
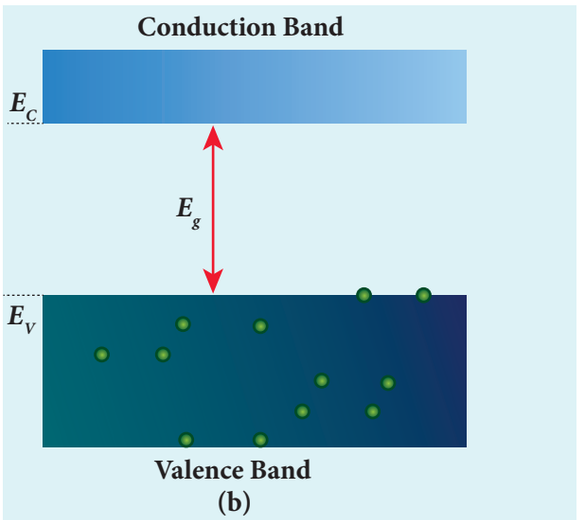
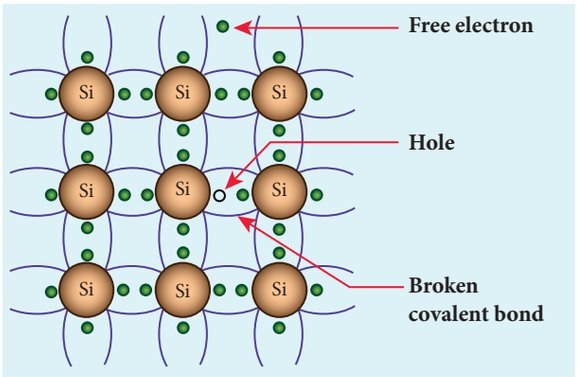


(a)

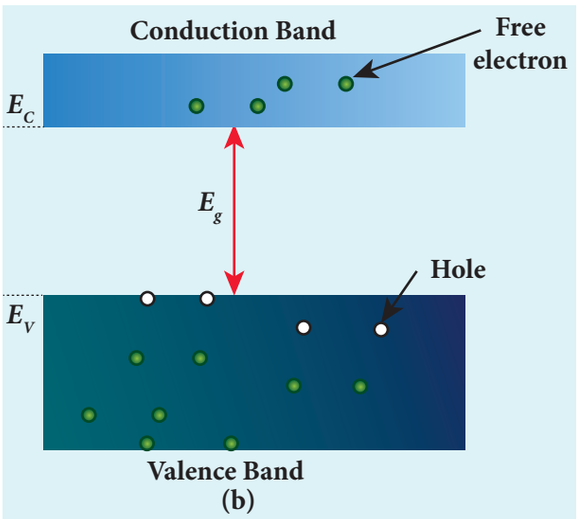


(b)

Figure 10.3 (a) Two dimensional crystal lattice of silicon (b) Valence band and conduction band of intrinsic semiconductor



(a)



(b)

Figure 10.4 (a) The presence of free electron, hole and broken covalent bond in the intrinsic silicon crystal (b) Presence of electrons in the conduction band and holes in the valence band at room temperature

presence of any other foreign atom in the crystal lattice. The silicon lattice is shown in Figure 10.3(a). Each silicon atom has four electrons in the outermost orbit and is covalently bonded with four neighbouring atoms to form the lattice. The band diagram for this case is shown in Figure 10.3(b).

A small increase in temperature is sufficient enough to break some of the covalent bonds and release the electrons free from the lattice (10.4(a)). As a result, some states in the valence band become empty and the same number of states in the conduction band will be occupied by electrons as shown in Figure 10.4(b). The vacancies produced in the valence band are called holes. As the holes are deficiency of electrons, they are treated to possess positive charges. Hence, electrons and holes are the two charge carriers in semiconductors.

In intrinsic semiconductors, the number of electrons in the conduction band is equal to the number of holes in the valence band. The electrical conduction is due to the electrons in the conduction band and holes in the valence band. The corresponding currents are represented as I_e and I_h respectively.

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